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Patient Appointment & Medical Record Management System

A large, abstract teal graphic in the bottom right corner, composed of several overlapping, rounded, organic shapes in different shades of teal and dark teal.

Domain Overview

Healthcare domain managing interactions between patients, medical doctors, and appointment scheduling, enforcing strict validation rules to maintain healthcare data integrity, secure communications, and proper operational status flows.

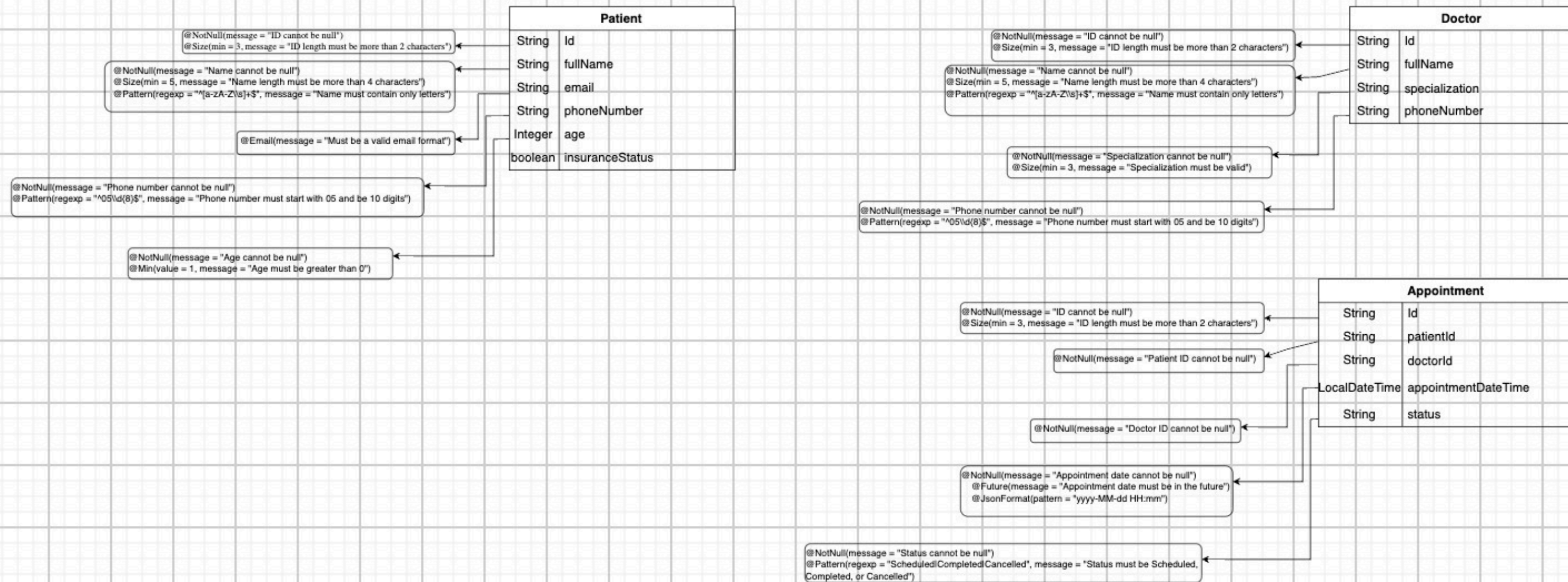


System Components

- Patient: Manages patient personal information, contact details, identification, insurance status, and medical eligibility.
- Doctor: Manages medical staff profiles, professional specializations, and contact details.
- Appointment: Manages scheduling links between patients and doctors, tracking dates, times, and sequential status workflows.



System UML Diagram



Model 1: Patient

```
test Version control
Patient.java Doctor.java Appointment.java
1 package com.waleed.test.Entity;
2
3 import jakarta.validation.constraints.*;
4 import lombok.AllArgsConstructor;
5 import lombok.Data;
6 import lombok.NoArgsConstructor;
7
8 @Data no usages
9 @AllArgsConstructor
10 @NoArgsConstructor
11 public class Patient {
12
13     @NotNull(message = "ID cannot be null")
14     @Size(min = 3, message = "ID length must be more than 2 characters")
15     private String id;
16
17     @NotNull(message = "Name cannot be null")
18     @Size(min = 5, message = "Name length must be more than 4 characters")
19     @Pattern(regexp = "[a-zA-Z\\s]+", message = "Name must contain only letters")
20     private String fullName;
21
22     @Email(message = "Must be a valid email format")
23     private String email;
24
25     @NotNull(message = "Phone number cannot be null")
26     @Pattern(regexp = "05\\d{8}", message = "Phone number must start with 05 and be 10 digits")
27     private String phoneNumber;
28
29     @NotNull(message = "Age cannot be null")
30     @Min(value = 1, message = "Age must be greater than 0")
31     private Integer age;
32
33     @AssertFalse(message = "Insurance status must be initially false")
34     private boolean insuranceStatus = false;
35 }
```

- ID:
 - Cannot be null.
 - Length must be more than 2 characters.
- Full Name:
 - Cannot be null.
 - Length must be more than 4 characters.
 - Must contain only characters (no numbers).
- Email:
 - Must be a valid email format.
- Phone Number:
 - Cannot be null.
 - Must start with "05".
 - Must consist of exactly 10 digits.
- Age:
 - Cannot be null.
 - Must be a number.
 - Must be greater than 0.
- insuranceStatus:
 - Must be initially set to false.

Model 2: Doctor

```
test Version control
Patient.java Doctor.java Appointment.java
1 package com.waleed.test.Entity;
2
3 import jakarta.validation.constraints.*;
4 import lombok.AllArgsConstructor;
5 import lombok.Data;
6 import lombok.NoArgsConstructor;
7
8 @Data no usages
9 @AllArgsConstructor
10 @NoArgsConstructor
11 public class Doctor {
12
13     @NotNull(message = "ID cannot be null")
14     @Size(min = 3, message = "ID length must be more than 2 characters")
15     private String id;
16
17     @NotNull(message = "Name cannot be null")
18     @Size(min = 5, message = "Name length must be more than 4 characters")
19     private String fullName;
20
21     @NotNull(message = "Specialization cannot be null")
22     @Size(min = 3, message = "Specialization must be valid")
23     private String specialization;
24
25     @NotNull(message = "Phone number cannot be null")
26     @Pattern(regexp = "^05\\d{8}$", message = "Phone number must start with 05 and be 10 digits")
27     private String phoneNumber;
28 }
```

- ID:
 - Cannot be null.
 - Length must be more than 2 characters.
- Full Name:
 - Cannot be null.
 - Length must be more than 4 characters.
- Specialization:
 - Cannot be null.
 - Length must be more than 2 characters.
- Phone Number:
 - Cannot be null.
 - Must start with "05".
 - Must consist of exactly 10 digits.

Model 3: Appointment

```
test Version control
Patient.java Doctor.java Appointment.java x
1 package com.waleed.test.Entity;
2
3 import com.fasterxml.jackson.annotation.JsonFormat;
4 import jakarta.validation.constraints.*;
5 import lombok.AllArgsConstructor;
6 import lombok.Data;
7 import lombok.NoArgsConstructor;
8 import java.time.LocalDateTime;
9
10 @Data no usages
11 @AllArgsConstructor
12 @NoArgsConstructor
13 public class Appointment {
14
15     @NotNull(message = "ID cannot be null")
16     @Size(min = 3, message = "ID length must be more than 2 characters")
17     private String id;
18
19     @NotNull(message = "Patient ID cannot be null")
20     private String patientId;
21
22     @NotNull(message = "Doctor ID cannot be null")
23     private String doctorId;
24
25     @NotNull(message = "Appointment date cannot be null")
26     @Future(message = "Appointment date must be in the future")
27     @JsonFormat(pattern = "yyyy-MM-dd HH:mm")
28     private LocalDateTime appointmentDateTime;
29
30     @NotNull(message = "Status cannot be null")
31     @Pattern(regexp = "Scheduled|Completed|Cancelled", message = "Status must be Scheduled, Completed, or Cancelled")
32     private String status;
33 }
```

- ID:
 - Cannot be null.
 - Length must be more than 2 characters.
- Patient ID:
 - Cannot be null.
- Doctor ID:
 - Cannot be null.
- Appointment Date & Time:
 - Cannot be null.
 - Must be a date in the future.
- Status:
 - Cannot be null.
 - Must be either "Scheduled", "Completed", or "Cancelled" only.